System Analysis & Design

Unit 1 – Solved Answer Key

Course Code: DSC-C-BCA-351T | Covers: Feasibility Study, Requirement Modeling (Fact-Finding Techniques), Data Flow Diagrams, Data Dictionary

Every answer below is built directly from the concepts already present in the Unit 1 notes.

A Long-Answer / Theory Questions

Q1 *Define fact-finding. Explain the Questionnaires and Survey method in detail.*

Answer

Fact-finding techniques are the methods used by systems analysts, researchers and project managers to gather data and requirements about a specific system under study (the "problem domain"). Their purpose is to discover the ground truth, define clear requirements, reduce project risk, and build stakeholder trust. The four standard fact-finding techniques are Interview, Document Review, Observation, and Questionnaires & Surveys.

Questionnaires and Surveys – Details

- Users are given a set of written questions; they fill in their answers themselves and return them to the systems analyst.
- Saves the analyst's time because every user does not need to be interviewed individually – especially useful when the time available for interviews is short.
- The analyst must clearly design and frame the questionnaire so it accurately captures the system's requirements and objectives.
- Works best when feedback is needed from a large or geographically spread group of users, and answers can be standardised or quantified easily.

> ■ ***Memory Tip:*** *"Q for Quantity" – reach for Questionnaires when you need opinions from a large Quantity of users, fast.*

Q2 *Define fact-finding. Explain the Interview method in detail.*

Answer

Fact-finding (see Q1 for the full definition) is the process of gathering data and requirements about the system under study using techniques such as Interview, Document Review, Observation, and Questionnaires/Surveys.

Interview Method – Details

An interview is a face-to-face interpersonal situation in which the interviewer (analyst) asks the interviewee questions designed to gather information about a problem. There are two types:

- **Unstructured Interview:** Conducted with only a general goal or subject in mind, with few or no specific pre-set questions. It uses **open-ended questions**, letting the user answer freely in their own words – useful in the early, exploratory stage of fact-finding.
- **Structured Interview:** Conducted using a **predefined set of questions**. It uses **close-ended questions** that limit the answer to specific choices or short, direct responses – useful for confirming or validating specific facts quickly.

■ ***Memory Tip:*** *Un-Structured = Un-limited (open-ended) answers; Structured = Specific/Short (close-ended) answers.*

Q3 *Define fact-finding. Explain the Document Review method in detail.*

Answer

Fact-finding (see Q1) covers the techniques analysts use to gather facts about the system under study.

Document Review – Details

The best way to analyse the existing system is to collect facts from existing documentation rather than relying only on human sources, which can be subjective. Documents the analyst typically reviews include:

- E-mails and customer complaints
- Suggestion-box notes
- Reports documenting the problem area and problem-performance reviews
- Samples of completed manual forms
- Samples of completed computerised forms/reports, and various flowcharts & diagrams
- Program documentation and user training manuals

■ ***Memory Tip:*** *"Documents don't lie" – always check the paper trail before relying purely on what people say.*

Q4 *Define fact-finding. Explain the Observation method in detail.*

Answer

Fact-finding (see Q1) covers the techniques analysts use to gather facts about the system under study.

Observation – Details

- The systems analyst personally participates in the organisation, studies the flow of documents, works directly with the existing system, and interacts with the actual users.
- It is especially useful when the analyst needs to capture the genuine user point of view, rather than a second-hand description of it.
- A specific technique called **work sampling** is used during observation – it lets the analyst understand how employees actually spend their working day, by sampling activity instead of watching continuously.

■ ***Memory Tip:*** *Observation = the analyst temporarily "becomes" an employee to see the real, unfiltered picture.*

Q5 *Explain feasibility study with all its types (Operational, Economic, Technical, Schedule).*

Answer

A **feasibility study** is conducted before committing money and time to a project, to check whether it is financially and technically worth doing. It is essentially a **reality check** – its goals are to prevent project failure and mitigate project risk. It always leads to one of three decisions: *go ahead*, *do not go ahead*, or *think again*.

Type	What It Checks
1. Operational	Will the system actually be used effectively once built? Depends on user acceptance, ease of operation, proper operator training, and how easily users can interface with the system.

2. Technical	Are the technical resources (hardware, software, network) and technically skilled manpower available to build, install and run the system? Inexperienced staff make the whole system insufficient.
3. Economic	Do the projected benefits outweigh the estimated total cost of ownership? Done as a cost-benefit analysis – can we afford the system, and what economic benefit (performance gain / cost cut) will it bring?
4. Schedule	Can the project be completed within an acceptable, agreed time-frame? This in turn depends on available manpower and economic conditions.

■ ***Memory Tip:*** *OTES – Operational, Technical, Economic, Schedule.*

Q7 *Explain the rules for developing a Data Flow Diagram (DFD).*

Answer

- Every **process** must have at least one input and one output.
- Every **data store** must have at least one data flow in and one data flow out.
- Data stored in the system must pass **through a process** – you cannot move data directly into or out of a store.
- Every process in the DFD must connect to **another process or a data store**.
- A data store must connect to a process (in, out, or both) – never directly to another data store or to an external entity.
- An **external entity** must connect to a process (in, out, or both) – never directly to a data store.
- A single data-flow arrow must only flow in **one direction**.

■ ***Memory Tip:*** *Process is the hub – every store or entity must touch a process; nothing talks to anything else directly.*

Q8 *Define data dictionary. Explain the advantages and disadvantages of a data dictionary.*

Answer

A **Data Dictionary** (also called a Data Definition Matrix) is a collection of data about data (**metadata**). It gives standard definitions of every data element used in the system – its meaning, allowable values, and logical/physical characteristics.

Advantages	Disadvantages
Consistency – keeps corporate data uniformly defined and accessed across the organisation, even across departmental boundaries.	Requires extra documentation effort and discipline to keep continuously updated as the system changes.
Clarity – makes data understandable and usable to both business users and developers, regardless of which division they belong to.	If poorly maintained, an outside user can misunderstand a data element's meaning/use, leading to an erroneous report and a wrong management decision.
Reusability – because data is defined consistently, components already designed, tested and approved can be reused in new development projects.	–

■ ***Memory Tip:*** *DD = Dictionary for Data – same job as a language dictionary, just for data elements.*

Q9 *Explain the rules for drawing a DFD.*

Answer

The core rules are the same as those covered under Q7 (every process needs an input and output, every data store needs a flow in and out, data must pass through a process, and a single flow only moves in one direction). In addition, while drawing a DFD, watch out for these classic mistakes:

- A **process with no incoming data flow** but only an outgoing flow – a process cannot generate output from nothing.
- A **data store connected directly to an external entity** – this lets the entity read/write the file without going through any process, bypassing all control/audit.
- A **data flow shown moving in both directions on one arrow** – use two separate arrows instead, one for each direction.

■ ***Memory Tip:*** *If a mistake breaks one of the rules from Q7, it shows up as one of these three classic errors.*

Q10 *Explain the rules for designing a data dictionary.*

Answer

- **Data Names:** Follow a consistent naming standard so every data element is recognisable across the project.
- **Definitions:** Write a precise definition that clearly distinguishes a data element from every other data element in the system.
- **Data Structures / Composites:** Define group or composite items by listing the data elements and sub-structures that make them up.
- **Keys & Relationships:** Identify entities, their occurrences, and how they relate through primary keys and foreign keys.
- **Transforms:** Document any process or operation that modifies the data.

■ ***Memory Tip:*** *Name it, Define it, Structure it, Key it, Transform it.*

Q11 *Write the steps to design a data dictionary.*

Answer

- **Step a – Identify the entity name:** Assign a proper, consistent, readable and standardised name.
- **Step b – Identify all its attributes:** Clearly define the data type and size for each attribute.
- **Step c – Identify relationships:** Mark primary keys and foreign keys to connect related entities.

■ ***Memory Tip:*** *Name → Attributes → Keys, in that exact order.*

B True / False Questions

Q6 *Processes or data stores can be shown in a context diagram.*

FALSE A context diagram (Level 0 DFD) is the highest-level, at-a-glance view of the system. It shows the **entire system as a single process**, connected only to its external entities. Data stores are an internal detail and only start appearing from Level 1 onward.

Q12 *Feasibility study is the first stage of the System Development Life Cycle (SDLC).*

TRUE Feasibility study is carried out before any design or development work begins – it is the initial planning-stage reality check that decides whether the project should proceed at all.

Q13 *System design is a process of collecting factual data.*

FALSE Collecting factual data, understanding processes, and recommending feasible suggestions is the definition of **System Analysis**, not System Design. System Design instead uses the analysis findings and user requirements to design the new system.

Q14 *Feasibility study precedes technical development and project implementation.*

TRUE By definition, a feasibility study is conducted before money and time are committed to development – it comes first, and the go/no-go decision it produces determines whether development even starts.

Q15 *Level 1 DFD provides an overview of the data entering and leaving the system.*

FALSE That description fits the **Context Diagram / Level 0 DFD**, which is the at-a-glance, single-process overview. **Level 1 DFD** instead provides a more detailed breakout of the pieces inside the Level 0 diagram.

Fill in the Blanks

Q.No	Statement	Answer
24	Data structures are also called _____________ data items.	**Group / Composite**
25	_________ is a collection of data about data.	**Data Dictionary**

C MCQ Answer Key (25 Questions)

Q.No	Question (Short)	Correct Answer
1	DFD stands for	**B. Data Flow Diagram**
2	Rounded rectangle / circle in a DFD represents	**C. Process**
3	NOT a valid component of a DFD	**C. Flowchart**
4	Level 0 DFD is also known as	**C. Context Diagram**
5	Symbol for an external source/destination of data	**B. Square / Rectangle**
6	Arrows in a DFD represent	**B. Data Flow**
7	Purpose of a Data Dictionary	**C. Provides definitions of data elements**
8	Usually NOT included in a Data Dictionary	**B. File size**
9	A Data Dictionary contains	**B. Information about data elements and structures**
10	Term most closely related to a Data Dictionary	**B. Documentation**
11	Primary purpose of a feasibility study	**B. To determine whether a project is viable**
12	NOT a common type of feasibility	**D. Political feasibility**
13	Technical feasibility deals with	**B. Availability of hardware and software**
14	Feasibility that checks if the system will function in the user's environment	**B. Operational**
15	Economic feasibility is also known as	**C. Cost-Benefit Analysis**
16	Feasibility type that examines compliance with laws/regulations	**B. Legal feasibility**
17	SDLC phase in which feasibility study is conducted	**C. Planning**
18	Outcome of a successful feasibility study	**C. Go / no-go decision for the project**
19	Processes or data stores can be shown in a context diagram	**False**
20	Feasibility study is the first stage of the SDLC	**True**
21	System design is a process of collecting factual data	**False**
22	Feasibility study precedes technical development & implementation	**True**
23	Level 1 DFD provides an overview of data entering/leaving the system	**False**
24	Data structures are also called _____ data items	**Group / Composite**
25	_____ is a collection of data about data	**Data Dictionary**

Note: Q19–Q23 and Q24–Q25 above repeat the same True/False and Fill-in-the-Blank items from Section B (questions 6, 12, 13, 14, 15, 24, 25) – they are listed again here only to keep this section a complete, single-page MCQ answer key.